

P3 — AI & Data Lead

RoadAssist Bharat · 70 days across 3 reviews Owns: 9 AI subsystems · model training and serving · on-device inference · feature store · voice and NLU · analytics pipeline. Backs up **P1**.

Your three standing rules 1. **Record the baseline before you train.** A model with no documented baseline cannot be said to work. 2. Every model ships with a **model card** and a **fallback**. No exceptions, including for models you are proud of. 3. No model is ever on the critical path for correctness or safety. **Models advise; rules decide.**

Your total is lower than the others on purpose. Model work is research-shaped and estimates are less reliable — the slack absorbs the experiments that fail, and some will.

The 9 AI Systems

#	System	Input → Output	Method	Runs on	Baseline to beat
1	Vehicle Diagnosis	Symptoms, DTC codes, history → ranked causes + severity + parts	Retrieval + gradient boosting; LLM only writes the explanation	Server + device	Rules on DTC lookup: top-3 = 0.62
2	Image Diagnosis	Photos → damage type, part, severity	MobileViT / EfficientNetV2, multi-label + severity head	Server + device	Non-expert labelling: top-3 = 0.55
3	Voice Assistant	Speech (8 languages) → intent + slots	Whisper fine-tuned on Indian speech → MuRIL intent classifier	Server; keyword on device	Keyword matching: 0.48
4	Predictive Maintenance	Telemetry + history → failure probability at 30/90 days	Survival analysis + LightGBM per component	Server, nightly	Fixed service interval: recall = 0.31
5	Mechanic Matching	Request + mechanic features → ranked list	LambdaMART ranker + separate ETA regressor	Server, <300 ms	Distance + rating sort: 0.41
6	Crash Detection	Accelerometer, gyroscope, GPS at 100 Hz → crash probability	1D CNN + GRU, INT8 quantised, ~1.5 MB	Device only	Peak-G threshold: 1 false alarm per 12 hours
7	Fraud Detection	Booking graph, payments, GPS traces → risk score + reasons	Graph features + isolation forest + gradient boosting	Server, async	Manual rules: precision 0.34
8	Government Analytics	Incidents, weather, road geometry → risk forecast, blackspots	Spatial clustering + temporal fusion model	Server, batch	Historical average by location
9	Offline AI Bundle	Symptoms, sensors → diagnosis + crash detection with no signal	Quantised and distilled versions of 1, 3 and 6	Device, ≤ 15 MB	The alternative is nothing

REVIEW 1 — Foundation · Weeks 1–5 · 13 days

Week	Task	Days
1	Data availability audit — what exists, what can be acquired, what must be synthetic	2
1–2	Feasibility spikes: crash detection from public sensor data; speech recognition on Indian-accented audio	2
2	Honest classification of all 9 systems: trainable now / trainable later / rules-only in v1	1
2	AI backlog (~35 stories), labelling plan and cost, evaluation protocol with baselines recorded	2
3	AI Gateway design: capability routing by model version, shadow mode, guardrails, fallback contract with P1	3
3–4	Feature store schema + label capture design (this is the highest-leverage decision of the whole review)	2
5	Fraud signal catalogue handed to P1; privacy-respecting device fingerprinting design	1

Deliverables

Data audit · feasibility report with the 9-system classification · labelling plan · evaluation protocol with recorded baselines · AI Gateway design · feature store schema · label taxonomy · model card template.

Your Review 1 demo — 5 minutes

Present the **honest classification**: which of the 9 systems are genuinely models in v1, which are rules wearing an AI label, and exactly what data would change that. Then run an end-to-end toy pipeline — data → train → registry → export → serve → prediction. The plumbing works before any real model exists. *This candour is what makes everything you claim later credible.*

Passed when

Every one of the 9 systems is classified honestly · every model has a **baseline recorded before training begins** · label capture is designed even though it will not be used for months · the training/serving feature parity test exists and fails when a feature is changed in only one place.

REVIEW 2 — Core Product · Weeks 6–12 · 33 days

Week	Task	Days
6–7	AI Gateway live with rule-based logic for all 8 capabilities — so P1 and P2 integrate on day one, months before any model exists	4
7	AI presentation contract with P2: confidence bands as plain language , not percentages	1
8–9	Vehicle Diagnosis model — retrieval + re-ranker + constrained LLM explanation	4
9	Image Diagnosis model — with augmentation for rain, night, motion blur, low-end cameras	3
9–10	Voice Assistant — speech recognition fine-tuning, intent + slots, code-switched Hinglish	3
10	Predictive Maintenance — survival analysis per component	3
10	Mechanic Matching — ranker + ETA regressor with an explicit fairness constraint	2
11	Crash Detection — the highest-stakes model in the project	3
11	Fraud Detection — graph features + anomaly detection	2
12	Model registry, shadow mode, one-config rollback	2
12	On-device bundle ≤ 15 MB — quantisation, distillation, benchmarking on 5 real devices	4
12	Vehicle-class rule packs (two-wheeler, auto, tractor) + EV diagnostics	2

Deliverables

AI Gateway with rules **and** models · 7 trained models · 7 model cards · evaluation harness · on-device bundle ≤ 15 MB · rule packs per vehicle class · EV diagnostics module.

Your Review 2 demo — 8 minutes

For each model: **the baseline, the model, the delta, and the failure modes**. Then a live diagnosis — photograph a real damaged part, get the fault, severity, parts list and a "do not drive" verdict in Hindi. Then unplug the models entirely and show the rules fallback keeping the product working. Finish in **airplane mode on a ₹8,000 phone**: describe a symptom, get a diagnosis in under 200 ms, nothing touching the network.

Passed when

Every model beats its recorded baseline on a held-out set · on-device inference under 120 ms · on-device bundle ≤ 15 MB · **measured false "safe to drive" rate is zero** · every model has a confidence threshold and a working fallback · quantisation accuracy loss reported, not assumed.

Rule packs are not a shortcut. Most Indian two-wheelers have no OBD port. For those vehicles, structured symptom questioning built with real mechanics is the *correct* engineering choice, not a fallback.

REVIEW 3 — Complete System · Weeks 13–18 · 24 days

Week	Task	Days
13	Crash detection to production quality — threshold tuning against real dogfooding data; severity model	3

Week	Task	Days
14	Blackspot detection (spatial clustering with significance testing); incident forecasting for government	3
15–16	Analytics — event taxonomy, ClickHouse schema, metric dictionary , anomaly alerting	5
16	Bias audit across all models — performance by vehicle class, language, region and device tier	2
16	Adversarial testing: prompt injection, adversarial images, sensor spoofing, GPS spoofing	2
17	Model serving in production; promotion pipeline with automatic rollback on metric regression	3
17–18	Inference optimisation: batching, caching, distillation, cost per million inferences	3
18	AI roadmap and the data flywheel — which model improves first, at what data volume	3

Deliverables

Production crash detection with measured false-positive rate · blackspot + forecasting models · ClickHouse analytics · metric dictionary · 9 finalised model cards with bias audits · model serving with auto-rollback · AI roadmap.

Your Review 3 demo — 6 minutes

Present the **real false-positive number** from 200 hours of driving. Then drive over a speed breaker on camera — nothing happens. That is the demo. Then run blackspot detection on a state's historical data and overlay the official blackspot list — show the overlap **and the new candidates you found**. Close by promoting a deliberately bad model and watching it roll back automatically in under 2 minutes.

Passed when

Crash false-positive rate ≤ 1 per 1,000 driving hours over **real** driving · the model raises a signal and can never auto-dispatch · every model has a bias audit · every metric has exactly one definition · a bad model rolls back automatically.

The pre-committed decision on crash detection. If the measured false-positive rate over ≥ 200 real driving hours does not meet the target, crash detection **ships disabled by default** and manual SOS carries the feature. This is decided now so it is not a judgement call under demo-day pressure.

Your Key Design Decisions (defend these to the panel)

Decision	Why
Rules ship first, models replace them	All 8 AI endpoints go live in Review 2 as rules. Models then upgrade them behind an unchanged contract. AI is never a blocker, and the rules remain the permanent production fallback.
The LLM never chooses the diagnosis	It explains a structured result selected by a classifier. This removes hallucination from the safety path entirely.
Crash detection runs on-device only	It must work with no connectivity — which is exactly the condition of a highway crash.
The matching model has a fairness constraint	A ranker optimised purely for acceptance concentrates all jobs on a few mechanics and destroys the marketplace. Job distribution is tracked as a first-class metric.
Fraud detection never auto-suspends	It queues for human review with mandatory reason codes. A false accusation destroys a mechanic's livelihood.
Precision over recall on maintenance alerts	A user who ignores three false warnings will ignore the real one.
Minimal MLOps stack	Tracking, versioning and plain containers. No orchestration platform — one person cannot operate that.